

MA323 : Lab 7 Report

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Q1

The following values of I_M were obtained after using the antithetic method.

M	I_M
50	1.99887
100	1.99802
500	1.99990
1000	2.00143
5000	2.00027
10000	2.00012
50000	1.99982
100000	2.00003

Q2

All the required value are summarised in the table given below. The ratio of the interval width is Lab7 / Lab6.

M	L6 Estimate	L7 Estimate	L6 CI	L7 CI	Ratio
100	1.96843	1.99802	(1.87996, 2.0569)	(1.99101, 2.00503)	0.07924
1000	2.01607	2.00143	(1.98891, 2.04323)	(1.99946, 2.00339)	0.07235
10000	1.99884	2.00012	(1.99025, 2.00744)	(1.99948, 2.00075)	0.07388
100000	1.99952	2.00003	(1.99678, 2.00226)	(1.99983, 2.00023)	0.07299

We can observe that both the methods are equally competent in getting the estimate of the required quantity (Actual value : 2). However, the method used in this lab's assignment is able to drastically reduce the width of the confidence interval (Close to 8% of the interval found in the Lab 6's assignment). This serves as a proof to the fact that this method is folds better than the previous method in terms of reducing the sample variance.